The Ontological Crisis of Modern Cosmology: From Phenomenology to Structural Physics

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Modern cosmology has achieved unprecedented mathematical precision and empirical success, yet remains ontologically empty. This work analyzes the fundamental contradictions of the ΛCDM model, demonstrates the incompatibility of standard cosmology with quantum mechanics at the level of basic principles, and substantiates the necessity of revising the ontology of spacetime. We show that dark energy and dark matter cannot be reconciled with Noether's theorem, and that the interpretation of cosmological redshift as a Doppler effect represents a historical choice rather than a logical necessity.

I. INTRODUCTION: THE PARADOX OF THE 21ST CENTURY

We live in a cosmology that is mathematically precise, empirically successful, and ontologically empty. This is truly the historical paradox of the 21st century. Dark energy in ΛCDM is not an explanation—it is a parameterization of ignorance. There exists no generally accepted ontological explanation of dark energy, and this is a fundamental problem, not a temporary gap in knowledge.

II. REDSHIFT: OBSERVATION VERSUS INTERPRETATION

A. Empirical Fact and Model Overlay

Redshift does not automatically equal accelerated expansion of space. This is an interpretation, not an observed fact. The observed fact is singular: spectral lines are shifted toward the red. Everything else constitutes a model overlay:

- Doppler interpretation
- Friedmann-Lemaître metric
- Cosmological constant Λ
- Concept of dark energy

B. Logical Inconsistency of the Expansion Concept

It does not logically follow from redshift that space is accelerating in its expansion. One cannot "expand nothing." If something expands, it must possess topology. Modern cosmology makes a strange methodological move: space is declared dynamic, but its ontology remains undefined.

What is actually employed is an implicit scheme of $\mathbb{R}^3$ "within" abstract meta-time without answering fundamental questions:

Is space embedded in anything?

Is it compact?

What global invariants does it possess?

This represents an implicit assumption of $\mathbb{R}^3 \subset \mathbb{R}^4$, though topology has never been explicitly postulated.

C. Why Topology Has Been Displaced

The question of topology is fundamental. If one honestly introduces topology, then:

- Global constraints appear
- Topological invariants emerge
- The freedom for arbitrary parameter fitting disappears
- The cosmological constant Λ ceases to be a free parameter

This explains why modern cosmology avoids the topological formulation of the question.

III. THE ONTOLOGICAL STRUCTURE OF THE UNIVERSE: THE 95% PROBLEM

The modern cosmological model asserts the following distribution of energy-matter in the Universe:

- 5% — baryonic matter
- 27% — dark matter (not directly detected)
- 68% — dark energy (ontologically undefined)

95% of the Universe's ontology lacks a direct observable carrier. This is unprecedented in the history of science. The claim that 68% of the Universe consists of "that which does not exist" is not an argument but a symptom of ontological crisis.

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A. Alternative Interpretation of Redshift

If the medium is topologically non-trivial, the scale is determined fractally, and phase and wavelength are connected to the global structure, then redshift may be a geometric rather than kinematic effect.

Modern cosmology does not consider this possibility because:

- It requires abandoning flat $\mathbb{R}^3$
- It necessitates introducing global topology
- It destroys ΛCDM as an "economical model"

B. The Observational Horizon Problem

If accelerated expansion is real, we should observe the disappearance of objects beyond the cosmological horizon. However, observationally: we do not see the disappearance of structures; on the contrary, the JWST telescope reveals ever deeper layers with increasing redshift, and structures do not simplify but become more complex.

This is consistent with a hierarchical or fractal structure of space, but poorly consistent with the model of a "young Universe" evolving from primordial chaos. ΛCDM is forced each time to post-hoc adjust the model: "structures must have formed faster." This is reactive rationalization, not predictive theory.

IV. THE HISTORICAL CONTINGENCY OF DOPPLER INTERPRETATION

A. Genesis of the Standard Interpretation

Why was redshift interpreted as a Doppler effect? This was a historical, not logical decision:

- In the early 20th century, the Doppler effect was already known
- No other working interpretations existed
- General relativity permits dynamic metrics but does not require redshift to be expansion
- After Hubble's discoveries, the linear z-d relationship was automatically interpreted as velocity

This was a choice by analogy, not a rigorous derivation. No one has proven that cosmological redshift is identical to kinematic Doppler effect. This is a postulate disguised as a derivation.

B. Institutional Displacement of Alternatives

Alternative interpretations existed and exist:

- Geometric
- Phase-based
- Medium-based
- Topological
- Wave-based

However, they have been displaced not experimentally but institutionally, because they require answering the question "what is space." ΛCDM does not pose this question, which ensures its methodological "flexibility."

V. ONTOLOGICAL GAPS IN STANDARD COSMOLOGY

If we remove the protective layers of mathematical formalism, standard cosmology does not explain:

- Entropy as the directionality of time (the second law of thermodynamics is simply postulated)
- Time as a physical entity (it remains a parameter, not an object of theory)
- Space (there is no metaspace, no ontology of expansion)
- Beginning (singularity is a mathematical breakdown, not a physical state)
- Dimensionality (why precisely 3+1 is not derived from first principles)
- Vacuum energy density (discrepancy with quantum predictions by 120 orders of magnitude)
- Ultraviolet divergences (removed by renormalization but not explained)
- Speed of light (postulate, not derivation)
- The reason for the existence of laws, not just their mathematical form

These are not "gaps in knowledge." These are ontological holes covered by computational success.

VI. FUNDAMENTAL INCOMPATIBILITY WITH QUANTUM MECHANICS

A. The Problem of Ontological Framework

The contradiction between ΛCDM cosmology and quantum mechanics lies not in specific parameters but in the ontological framework. Even if dark matter and dark energy are one day formally "discovered" or explained, they cannot be reconciled with quantum mechanics because the conflict is embedded in the very structure of the model.

1. ΛCDM Ontology

The ΛCDM model employs an ontology of $\mathbb{R}^3 \subset \mathbb{R}^4$:
Space is treated as substance
Time is an external parameter
Expansion is a physical process
Vacuum is a background structure
Energy is an addition to geometry
This is classical metaphysics dressed in mathematical formalism.

2. Requirements of Quantum Mechanics

Quantum mechanics presents fundamentally different requirements:

- Unitarity → information conservation → reversibility at the fundamental level
- Non-locality (in the sense of Bell's inequalities) → impossibility of classical causal structure in $\mathbb{R}^3$
- Absence of external time → time as evolution parameter is already an approximation
- Absence of absolute vacuum → "emptiness" is not background but dynamic structure of quantum state

B. Structural Incompatibility

ΛCDM in the framework of $\mathbb{R}^3 \subset \mathbb{R}^4$ requires:
External absolute time
Local geometry
Background vacuum
Additional entities (dark matter and dark energy)
Effective rather than fundamental unitarity
Even ideally selected parameters of dark matter and dark energy do not eliminate any of these conceptual conflicts. They mask phenomenology but do not change ontology.

New dark matter particles do not solve the time problem. Dark energy fields do not solve the unitarity problem. Modified gravity does not solve the non-locality and information conservation problems. These are modifications within a flawed architecture.

Conclusion: If cosmology permits fundamental non-unitarity through dark energy, expanding vacuum, or metric creation, it cannot be compatible with quantum mechanics regardless of what entities are added to it.

VII. RANDOMNESS AND CAUSALITY: A LOGICAL CONTRADICTION

A. The Ontological Status of Randomness

A random event cannot serve as the foundation of a stable cause-and-effect reality. If the opposite is claimed, explicit arguments are required. Cause-and-effect reality presupposes:

- Stable laws
- Repeatability
- Invariants
- Predictable connections

A random event by definition:

- Is not derived from law
- Contains no internal necessity
- Does not guarantee reproducibility

B. Logical Inconsistency of Random Beginning

Randomness can be an interpretation of unrecognized structure, but not its source. Otherwise a logical contradiction arises:

- From non-lawful → lawful
- From non-structural → structural
- From "without cause" → causality

Factually, there exists no theory that derives causality from randomness, time from an act "outside time," or laws from their absence.

Thesis: The foundation of reality cannot be an event. It must be structural, atemporal, topologically defined, admitting local fluctuations but not generated by them.

Consequently, the Big Bang as a random beginning cannot be the ontological foundation of reality.

C. Quantum Mechanics and Randomness

The assertion that "quantum mechanics is fundamentally random" is an incorrect formulation. Quantum theory asserts:

- Probabilistic measurement outcome
- With strictly deterministic wave function evolution

That is:

- The Schrödinger equation is not random
- The state space is not random
- Symmetries are not random
- Probability is not ontology but a measure

The distinction is critical: probability is a measure on state space; randomness is an assertion about the absence of cause.

Quantum mechanics uses probabilities but does not require ontological randomness. It is fully compatible with the thesis: the result has a cause, but the cause may be inaccessible to a local observer.

D. Randomness and Information Conservation

Fundamental randomness is logically incompatible with the law of information conservation in quantum mechanics.

The Law of Information Conservation

In quantum mechanics and quantum field theory, state evolution is determined by a unitary operator:

$$|\psi(t_2)\rangle = U(t_2, t_1)\,|\psi(t_1)\rangle, \qquad U^\dagger U = I. \tag{1}$$

Unitarity is equivalent to reversibility; reversibility is equivalent to information conservation. This means:

- Each state has a unique prehistory
- No information arises from nothing or disappears into nothing

This is a rigid mathematical requirement, not a philosophical choice.

Conflict with Ontological Randomness

To assert that randomness is ontological means:

- The result does not follow from the complete state of the system
- There is no complete set of causes
- The future is not encoded in the past even potentially

If fundamental randomness is real, unitarity is violated and the law of information conservation is false. However, all of quantum theory is built on unitarity, which is experimentally confirmed with the highest precision. Its rejection destroys the theory itself.

Syllogism:

- Quantum mechanics is unitary
- Unitarity is equivalent to information conservation
- Information conservation requires causal reversibility
- Ontological randomness violates reversibility
- Therefore, ontological randomness is incompatible with quantum mechanics

This is a structural conclusion, not an opinion. If quantum mechanics is correct, randomness cannot be fundamental. If randomness is fundamental—quantum mechanics does not exist.

Conclusion: Randomness is an absolutely contradictory principle relative to the law of information conservation in quantum mechanics. Randomness is an epistemic convention designating the boundary of understanding, not a source of being.

VIII. NOETHER'S THEOREM AS A CRITERION OF PHYSICALITY

A. Fundamental Significance of the Theorem

No physical model can be constructed outside Noether's theorem. This is not a matter of interpretation. Emmy Noether's theorem (1918) establishes a rigid equivalence:

- Symmetry $\Leftrightarrow$ conservation law
- Temporal symmetry $\Leftrightarrow$ energy conservation
- Spatial symmetry $\Leftrightarrow$ momentum conservation
- Phase symmetry $\Leftrightarrow$ probability conservation (unitarity)

Criterion: If a theory introduces an entity but cannot indicate the corresponding symmetry, it has stepped outside physics, even if the mathematical formalism appears correct.

B. The Problem of Dark Matter and Dark Energy

In the ΛCDM model:

Dark energy is not derived from symmetry

It is not connected to a transformation generator

It violates global energy conservation in an expanding background

Dark matter is postulated as "something" without derivation from a fundamental Lagrangian

It has no unique symmetry (except ad hoc introduced ones)

This is a violation of Noetherian logic, not simply a matter of experiment. Even if one formally "fits" dark matter and dark energy into equations, physics in the strict sense will not result.

C. Impossibility of Detection

Symmetry is a property of structure. If there is no structure, there is no symmetry. Therefore:

- Decades of dark matter searches remain fruitless
- "Dark energy fields" have no corresponding generators
- New detectors search for ontological emptiness

Question: How can one detect that which does not exist as a physical entity?

Answer: Only by non-physical methods. And this is no longer science.

D. Structural Verdict

A physical theory must satisfy Noether's theorem. Noether's theorem requires symmetry for each conservation law. Dark matter and dark energy have no derived symmetries. Therefore, they are not fundamental physical entities.

The observed effect must be either geometric, phase-related, or complex (connected to the imaginary part of a generator), but not new matter or energy in the traditional sense.

Conclusion: Only one option remains—to reconsider the very structure of spacetime and evolution generators. Dark matter and dark energy represent an attempt to preserve phenomenology at the cost of destroying Noether's theorem. Such "physics" is impossible. This is a structural verdict on the framework, not the parameters.

IX. ALTERNATIVE ONTOLOGY: STRUCTURAL COMPLETENESS

A. The Full Moon Metaphor

Reality is always complete, like the full moon. This is an ontological metaphor:

- The moon is always full → this is ontology, complete structure
- Moon phases → this is epistemology, effect of observer position
- Disappearance and growth → illusion arising from incomplete section of the whole

This is not poetry but a precise distinction between global topology and local observational slice that exists in physics.

B. Observer within Structure

An observer arising within structure is always located in the complete present, while past and future are conceptual reconstructions, not ontological regions.

This is consistent with three independent lines of thought:

- Buddhist Ontology (Gautama Shakyamuni tradition):
- There is no "arising of the world" as an event
- There is interdependent manifestation (pratītyasamutpāda)
- Time is not a container but a relation
- Modern Physics:
- The observer is always local
- "Beginning" is an extrapolation of equations, not an experienced fact
- No observer has ever observed the "arising of reality"
- Logical Derivation:
- No observer experiences a "beginning"
- No observer is located "in past or future"
- Wholeness is always given completely but seen partially

The present is always complete—this is not consolation or mysticism. This is a rigorous conclusion. Reality does not become complete in time—it is always complete, and time is the way the observer learns to see this.

C. Gautama Shakyamuni's Position as Rigorous Ontology

"Reality is empty of what is not, and not empty of what is."

"Empty" (śūnyatā) — has no inherent existence (svabhāva)

"Not empty" — is completely interdependent (pratītyasamutpāda)

Consequently:

- Randomness as something without cause has no ontological status
- Causality is not optional but the only possible mode of existence

This is not belief. This is exhaustive logic.
Two Levels:

- Absolute Level (Ontology):
- Nothing arises without cause
- There are no events "from nowhere"
- There are no self-existent acts
- Relative Level (Epistemology):
- Complex causal chains appear random
- A limited observer does not see the entire structure
- "Randomness" is a marker of ignorance, not a property of the world

This is fully consistent with complex systems physics, probability theory, quantum mechanics (in correct interpretation), and rigorous logic.

X. HISTORICAL PARALLELS

A. The Geocentric System

"The model works" $\neq$ "the model correctly describes reality."

The geocentric system worked perfectly for centuries. It was:

- Computationally convenient
- Phenomenologically adequate
- Ontologically false

The same is happening with ΛCDM: it is engineered useful, but this is not an argument for its ontological truth.

B. Institutional Inertia

Modern physics teaches computational loyalty, not ontological thinking. Questions such as:
What exactly is expanding?
What is space?
Why does redshift arise at all?
— are declared "philosophy" and removed from scientific discourse.

For most researchers, it is safer to work in modes where any failure can be attributed to "parameters," "systematic errors," or "lack of data." In the mainstream, operationalism prevails: "as long as it calculates and matches."

XI. CONCLUSION: THE NECESSITY OF ONTOLOGICAL TURN

A. Diagnosis

If we remove the protective layers of formulas, ignorance in modern physics is not local—it is total, but disguised as "we almost understand everything."

Historically, science could exist with ignorance when it was limited. Now ignorance has become systemic, structural, but is presented as a set of particular problems.

B. The Way Forward

The solution is not in seeking the "correct theory of everything" within the current paradigm. What is necessary is a return to ontology through recognition:

- Reality is structured, not chaotic
- Time is not a parameter but a relation
- Space is not emptiness but a form of connectivity
- Humans are participants in structure, not observers outside the system

C. Ethical Dimension

Without ontological understanding, any science becomes a tool without ethics. If we do not understand where we are and what this reality is, do we have the right to control it and each other?

This is not "philosophy for beauty." This is a question of civilization survival.

D. Requirements for Future Theory

A future physical theory must:

- Explicitly define the topology of spacetime
- Derive symmetries for all conserved quantities (Noether requirement)
- Ensure consistency with quantum mechanical unitarity
- Explain redshift from structural rather than kinematic principles
- Eliminate the ontological emptiness of "dark entities"
- Reconcile global causality with local observability

Recognition of ΛCDM's ontological crisis means not "correcting the model" but abandoning it as an ontological framework while preserving its phenomenological utility as a computational tool.

Key Thesis of the Work: Cosmology constructed in the ontology of $\mathbb{R}^3 \subset \mathbb{R}^4$ with fundamental expansion is incompatible with quantum mechanics by definition. The very idea of expanding substance destroys unitarity and causal integrity. Until the ontology of spacetime is reconsidered, no "dark entities" can reconcile cosmology with quantum mechanics.

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